

Image Enhancement using Fourier Transform

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Abstract— Image enhancement is a critical step in digital image processing, aimed at improving the visual appearance or making images more suitable for analysis. This paper explores the application of the Fourier Transform (FT) for image enhancement. The FT is a powerful mathematical tool that decomposes an image into its sinusoidal components. By manipulating the frequency domain, significant improvements in image quality, edge sharpness, and noise reduction can be achieved. The research presents various techniques, implementation steps, and the benefits of frequency-based image enhancement, particularly focusing on the removal of periodic noise and enhancement of image details.



INDEX TERMS— KEYWORDS

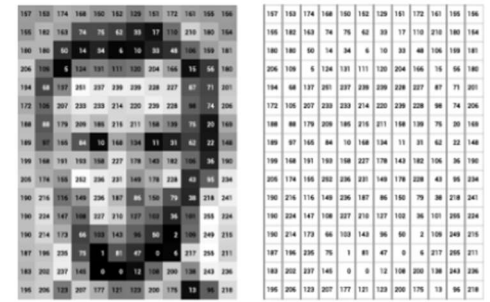
Image Enhancement, Fourier Transform, Frequency Domain, Noise Removal, Edge Detection.

I. INTRODUCTION

Digital image processing involves techniques to improve image quality for better human perception or machine interpretation. Image enhancement focuses on highlighting certain features of interest in an image or suppressing unwanted distortions. Spatial domain methods modify pixel values directly, while frequency domain methods, such as those using the Fourier Transform, work on the image's frequency content, offering unique advantages in processing periodic patterns and reducing noise.

II Image Storage in the Spatial Domain

Digital images are commonly stored in the **spatial domain**, where each pixel's position directly corresponds to its intensity value. An image is represented as a 2D array $f(x,y)$, where x and y represent the spatial coordinates and the value at each location signifies the brightness or color of that pixel. Spatial domain methods manipulate these pixel values directly to perform operations like contrast enhancement, sharpening, and blurring.



III FOURIER TRANSFORM IN IMAGE PROCESSING

The Fourier Transform converts spatial domain images into the frequency domain, allowing us to analyze the image content based on frequency components. High frequencies represent edges and fine details, while low frequencies indicate smooth variations like background lighting **Fig. 1**.

3.1 Discrete Fourier Transform (DFT)

The DFT is used for image processing in computers and is given by:

- Forward 2D Discrete Fourier Transform (DFT):

$$F(u, v) = \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} f(x, y) \cdot e^{-j2\pi(\frac{xu}{M} + \frac{yv}{N})}$$

- Inverse 2D Discrete Fourier Transform (IDFT):

$$f(x, y) = \frac{1}{MN} \sum_{u=0}^{M-1} \sum_{v=0}^{N-1} F(u, v) \cdot e^{j2\pi(\frac{xu}{M} + \frac{yv}{N})}$$

- Filtering in Frequency Domain:

$$G(u, v) = H(u, v) \cdot F(u, v)$$

$$g(x, y) = \text{IDFT}\{G(u, v)\}$$

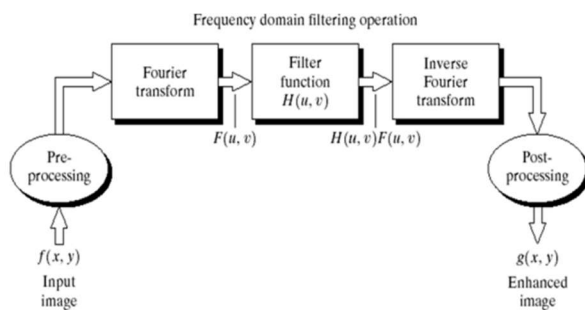
- Where:

- $f(x, y)$: Input image in spatial domain
- $F(u, v)$: Image in frequency domain
- $H(u, v)$: Filter function
- $G(u, v)$: Filtered frequency spectrum
- $g(x, y)$: Enhanced image in spatial domain

3.2 Fast Fourier Transform (FFT)

FFT is an efficient implementation of DFT, significantly reducing computation time, which is critical for real-time processing.

Architecture



A. Abbreviations and Acronyms

Abbreviation	Full Form
DFT	Discrete Fourier Transform

Abbreviation

Full Form

FFT	Fast Fourier Transform
FT	Fourier Transform
IFT	Inverse Fourier Transform
LLIE	Low-Light Image Enhancement
SNR	Signal-to-Noise Ratio
CNN	Convolutional Neural Network
GAN	Generative Adversarial Network

III. MATH

Step-by-Step Manual 2D DFT (2x2 Image)

Let's take this 2x2 grayscale image:

$$f(x, y) = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$$

★ Formula for 2D DFT:

$$F(u, v) = \sum_{x=0}^1 \sum_{y=0}^1 f(x, y) \cdot e^{-j2\pi(\frac{xu}{2} + \frac{yv}{2})}$$

We will compute $F(u, v)$ for $u = 0, 1, v = 0, 1$. So, there are 4 outputs: $F(0, 0), F(0, 1), F(1, 0), F(1, 1)$

✓ Compute Each Term

- $F(0, 0)$

$$F(0, 0) = \sum_{x=0}^1 \sum_{y=0}^1 f(x, y) \cdot e^{-j2\pi(0+0)} = \sum f(x, y)$$

$$= 1 + 2 + 3 + 4 = \boxed{10}$$

♦ $F(0,1)$

$$F(0,1) = \sum_{x=0}^1 \sum_{y=0}^1 f(x,y) \cdot e^{-j2\pi(0+\frac{y}{2})}$$

$$= f(0,0) \cdot e^0 + f(0,1) \cdot e^{-j\pi} + f(1,0) \cdot e^0 + f(1,1) \cdot e^{-j\pi}$$

$$= 1(1) + 2(-1) + 3(1) + 4(-1) = 1 - 2 + 3 - 4 = \boxed{-2}$$

♦ $F(1,0)$

$$F(1,0) = \sum_{x=0}^1 \sum_{y=0}^1 f(x,y) \cdot e^{-j2\pi(\frac{x}{2}+0)}$$

$$= f(0,0) \cdot e^0 + f(0,1) \cdot e^0 + f(1,0) \cdot e^{-j\pi} + f(1,1) \cdot e^{-j\pi}$$

$$= 1 + 2 + 3(-1) + 4(-1) = 1 + 2 - 3 - 4 = \boxed{-4}$$

♦ $F(1,1)$

$$F(1,1) = \sum_{x=0}^1 \sum_{y=0}^1 f(x,y) \cdot e^{-j2\pi(\frac{x}{2}+\frac{y}{2})}$$

Let's compute:

- $e^{-j2\pi(\frac{0}{2}+\frac{0}{2})} = 1$
- $e^{-j2\pi(\frac{0}{2}+\frac{1}{2})} = e^{-j\pi} = -1$
- $e^{-j2\pi(\frac{1}{2}+\frac{0}{2})} = e^{-j\pi} = -1$
- $e^{-j2\pi(\frac{1}{2}+\frac{1}{2})} = e^{-j2\pi} = 1$

$$F(1,1) = 1(1) + 2(-1) + 3(-1) + 4(1) = 1 - 2 - 3 + 4 = \boxed{0}$$

$$F(u,v) = \begin{bmatrix} 10 & -2 \\ -4 & 0 \end{bmatrix}$$

This is the image in the **frequency domain**!

B. Algorithms

1. To analyze the frequency components of a grayscale image, the image is first converted to the frequency domain using the **2D Discrete Fourier Transform (DFT)**. This process transforms the image from the **spatial domain**—where pixel intensities are directly interpreted—to the **frequency domain**, where each point represents a particular frequency contained in the spatial image.

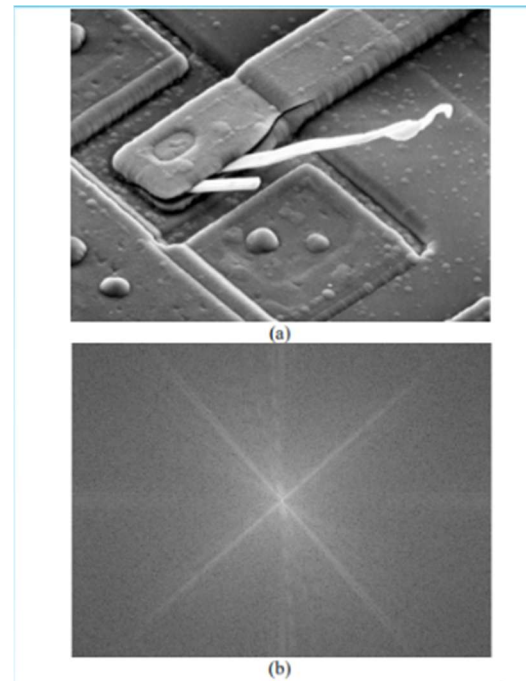
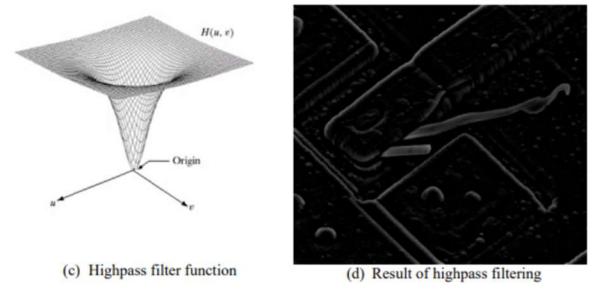


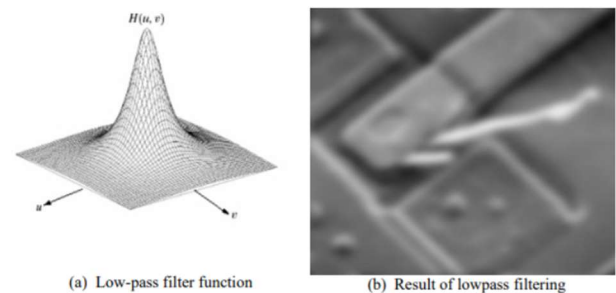
Figure 6.1 (a) Gray image. (b) Centered Fourier spectrum of (a)

2. Applied high Pass filter



A **high-pass filter (HPF)** allows high-frequency components to pass through while attenuating or blocking low-frequency components such as gradual changes in intensity. This makes the filter particularly effective in **edge enhancement and highlighting fine details** in an image.

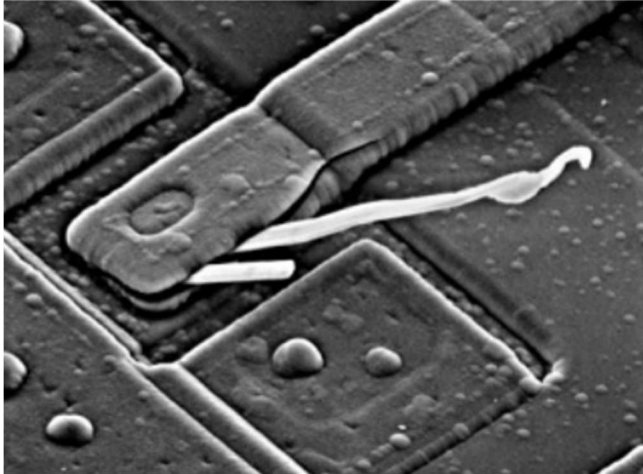
3. Can Also Apply low pass filter



A **low-pass filter (LPF)** allows **low-frequency components** to pass through while attenuating the **high-frequency components** of the image. This

operation smoothens the image by reducing sharp transitions such as noise and edges.

4. Enhancing Filter Response by Adding a Constant (0.75)



To refine the performance of both **high-pass** and **low-pass filters**, a constant value (e.g., **0.75**) can be added to the filter function. This technique is used to **tune the filter's strength** or to **avoid extreme suppression** of certain frequency components. Such adjustments help preserve important visual details that might otherwise be lost.

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